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Plastic to apparel: an analysis of sustainable purchasing intention using a machine learning ensemble

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The use of plastics has become a significant component in maintaining the convenience and suitability of modern lifestyles; however, a vast majority of the million tons of plastic manufactured each year ends up in landfills, contributing to plastic pollution. With this, the fashion industry has capitalized to create recycled products. Despite the proliferation and continued presence of recycled and upcycle products, there still is a significant gap in the sustainable purchasing behavior of consumers. This study aimed to identify, analyze, and forecast the variables influencing consumers' behavioral intention toward purchasing apparel made from plastic. This paper established the Sustainability Theory of Planned Behavior model to determine the purchase intentions of Filipino customers while purchasing clothing made of recycled plastic. A total of 500 valid respondents were gathered to evaluate factors: Perceived Economic Concern, Perceived Environmental Concern, Perceived Authority Support, Subjective Norm, Attitude, Perceived Behavioral Control, Customer Perceived Value, and Behavioral Intention. To analyze the data, the study utilized machine learning methods, such as Random Forest Classifier (RFC) and Artificial Neural Network (ANN). Data pre-processing using feature selection and correlation analysis was conducted to validate the available data, performed data cleaning process, and data aggregation. Several iterative processes were employed to generate the optimum classification model—obtaining a 92% accuracy for RFC and 91% for ANN at 150 epochs under 30 hidden layer nodes. With low error rates, the findings revealed that customer perceived value and perceived behavioral control were the primary factors influencing consumers' behavioral intentions toward purchasing sustainable clothing. This study emphasized the consideration of these factors when planning marketing strategies and initiatives to promote sustainable apparel.

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Introduction

The use of plastics has become a significant component in maintaining the convenience and suitability of modern lifestyles. Considering its adaptability and remarkable cost-performance ratio over the past several years, it has encompassed everything from everyday life to commercial manufacturing (Chang et al. 2019). Since then, majority of our daily activities have incorporated the usage of plastics—ranging from various food and beverage, cosmetics, toiletries, pharmaceuticals, and other products. These industries require packaging for their finished products in preparation of its safe and effective distribution to customers (Evođe et al. 2021).

Shams et al. (2021) discussed how the overwhelming bulk of the annual one million tons of plastic production, including items like water containers, bags, food containers, gloves, and cups, are discarded after every single use. In addition, the study by Zhang et al. (2022) stated that plastic pollution poses dangerous health repercussions for both humans and marine species. In areas where industrial wastes like plastics, rubber, and textiles are frequently burned, fumes and the release of toxic substances into the air, producing unpleasant odors from waste materials—contributing significantly to air pollution. Furthermore, a study conducted by Awoyera and Adesina (2020) mentioned that out of countless tons of plastic garbage, only 7% is recycled, approximately 8% is burned, and the remaining is landfilled. To which, consequences of the increasing price and energy related to the landfilling process led to water pollution caused by waste discarded into bodies of water.

Tiseo (2023) posited how the Pasig River in the Philippines releases over 63,000 metric tons of plastic debris into the ocean every year (Fig. 1). The data in 2019 shows that the Pasig River was estimated to have contributed 6.43 percent of all river-derived ocean plastics, making the Philippines the world’s heaviest contributor of plastic-polluting rivers.

OECD (2022) reported that 22% of plastic waste was improperly handled and not collected, 19% was burned, 49% ended up in landfills, and only 9% was recycled (Fig. 2). The growth in emerging economies has caused the use of plastic to triple over the previous 30 years. In recent times, plastics have been responsible for 3.4% of global greenhouse gas emissions, with this trend observed between 2000 and 2019, there was a double increase in global plastics manufacturing to 460 million tons.

In the fast fashion industry, one of their objectives is to manufacture and dispose of clothing rapidly; it pertains to the

speed at which the manufactured products are produced and then discarded, aiming for instant consumption. This market became a globalized industry, utilizing cheap labor and materials all around the world (Zhang et al. 2021). This led to an increase in non-recyclable waste in landfills (Gomes de Oliveira et al. 2022). Different studies (Niinimäki et al. 2020; Brewer 2019) explained that fast fashion is now the industry with the second-highest pollutant emissions at 10%. It was explained that large amounts of raw materials are needed for manufacturing fashion, which produces a substantial amount of waste that leaves a considerable carbon footprint and produces a significant level of pollution (Grazzini et al. 2021).

Nowadays, clothing companies are widely known for putting a lot more effort into producing environment-friendly apparel that focuses on sustainability. H&M, Adidas, and Nike are among the many popular international clothing companies that have committed to driving advancements toward an improved fashion future. According to H&M Group (2023), their resources are aimed to be either 100% recycled or obtained through more environmentally friendly means by 2023, with 30% recycled materials by 2025. H&M also noted the use of recycled plastic, derived from sources like PET plastic drinking bottles, plastic bags, shampoo containers, and various other plastic packaging is one of its most often obtained components, in which several of their well-known accessories are made with recycled plastic. This approach could prevent further damage to the environment. In addition, Adidas engages in a variety of environmental initiatives. One of which is using recycled plastic in manufacturing their products, which is a cornerstone of its commitment to reducing plastic waste and reducing and preventing pollution in the world’s oceans. The collaboration between Adidas and Parley is one of the brand’s sustainability initiatives; Adidas gave the plastic waste from beaches and coastal towns a new life as an Adidas x Parley product by intercepting the waste before it reached the ocean. Another project is initiated by Nike, starting with the ‘Move to Zero’ program. The journey aims to reduce waste and carbon emissions to safeguard the future of the sport. One of the materials they utilize is recycled polyester, which is created by shredding plastic bottles, turning them into granules, and then twisting the granules into high-quality yarn. Nike is currently using recycled polyester made from shredded plastic to lessen waste, approximately as much as 30% in comparison to newly produced polyester, and it helps keep 1 billion plastic

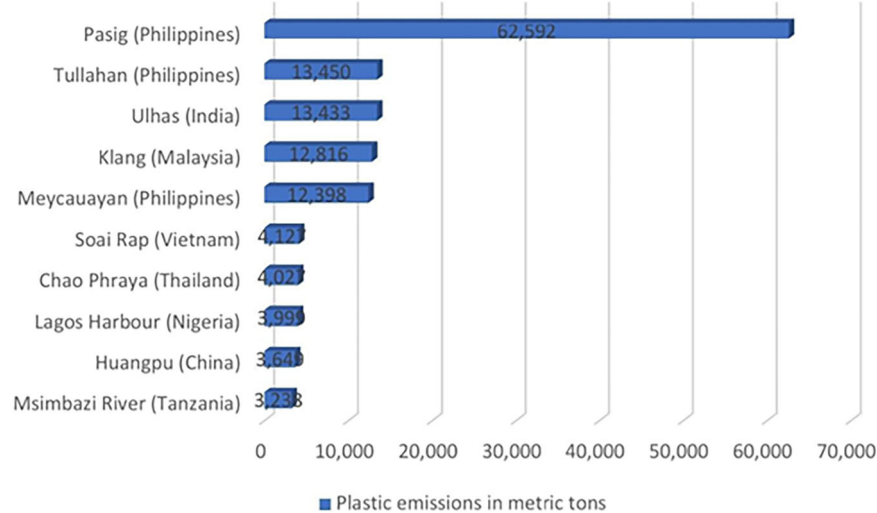


Fig. 1 Annual report of plastic waste emissions from selected rivers globally to the ocean as of 2019.

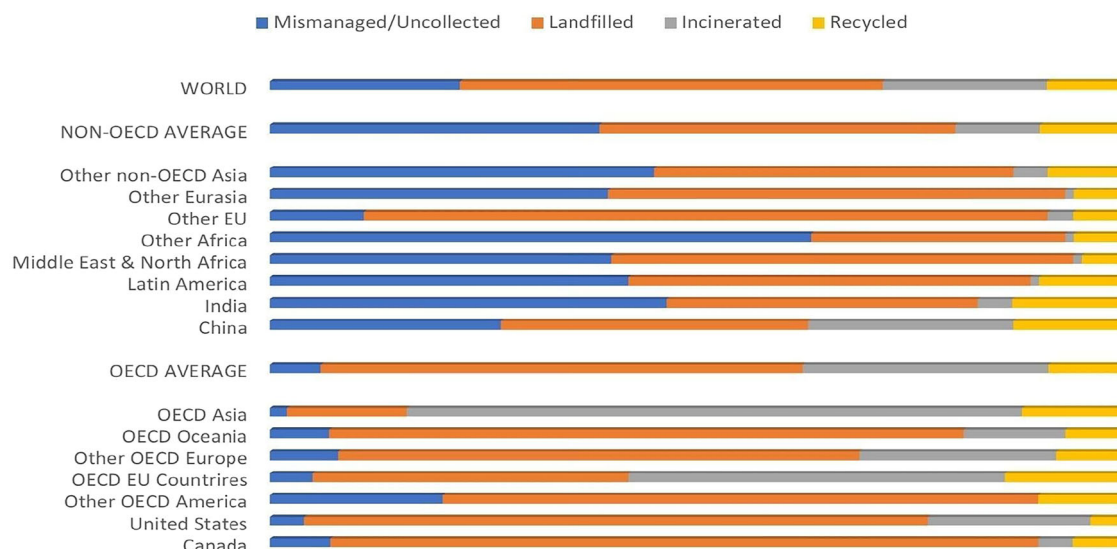


Fig. 2 OECD report of the management of plastic pollution growth globally, as of 2019.

bottles out of landfills and rivers and streams each year (Nike Sustainability 2024).

In recent years, much significant research has focused on customers' attitudes and behavior regarding sustainable fashion products by exploring the importance of sustainability within the fashion sector (Grazzini et al. 2021). Much research on environment-friendly clothing has looked at the potential benefits of eco-conscious product development approaches (Fung et al. 2021; Provin et al. 2021), how customers engage with sustainable fashion brands on social media platforms (Testa et al. 2021), along with how consumers perceive, their attitudes toward, and their willingness to invest in sustainable fashion products, and the factors that affect these behaviors (Grazzini et al. 2021; Nike Sustainability 2024).

Presented in Table 1 are the summarized key related studies alongside the limitations and need for future works.

Nguyen et al. (2020) conducted a fashion-focused survey in Vietnam and found that 86% of the respondents were aware of the potential to incorporate recycled plastic waste into the fashion industry. Kim et al. (2021) highlighted that concerns about the aesthetic aspects of clothing made from recycled materials could relate to worries about how well these garments blend with the consumer's existing wardrobe, their ability to align with the consumer's desired self-image, and their comfort in terms of size. As a result, consumers might delay or choose not to buy products made from recycled plastic materials (such as clothing) due to aesthetic risk (Kim et al. 2021; Testa et al. 2021). These studies demonstrated that aside from sustainability domains, people's behavior encompasses behavioral intention and actual behavior (Park and Lin 2020; Kuah and Wang 2020; Nguyen et al. 2020; Kim et al. 2021; German et al. 2022a).

In relation, Polyportis et al. (2022) mentioned that consumers who experience positive emotions as a byproduct of their efforts to lessen environmental harm are among the effective responses. Customers perceive that selecting and valuing products crafted from recycled materials would evoke positive and comforting emotions, such as pride as a result of their contribution to a better world (Adigüzel and Donato 2021). Moreover, Magnier et al. (2019) referred to expected moral awareness, which is characterized as a consumer's hopes regarding the way the goods would make him or her feel from an ethical perspective. Tezer and Bodur (2019) referred to the "warm glow" sentiments that come with just utilizing eco-friendly products, like those made from recycled materials; it was also highlighted that an increase in

the level of social value placed on customers as individuals, which contributes to these warm glow feelings, which improves how much you enjoy the accompanying consumption experience. Thus, perceived customer value is evident among sustainable behaviors, which should be considered when assessing consumer behavior (German et al. 2022a).

Despite many studies exploring sustainable behaviors, it could be deduced that this recent advancement in apparel, sustainable practices, and consumption has still been underexplored—implicating a research gap in the current trend of apparels. The holistic measurement of sustainable behavior should be investigated to assess the behavioral intentions of consumers. The novelty of this study lies with the sustainability domains, which were one of the factors which was adopted in this study (German et al. 2022a). Under the sustainability domains, five factors are being considered such as the human, environmental, economic, productivity, and social aspects (Hajishirzi et al. 2022). On the other hand, an established theory in the field of behavior, known as the Theory of Planned Behavior (TPB), has been accessible and extensively contemplated. It measures a person's behavioral characteristics, such as the social aspects that pertain to social ties and structures that promote stability and stability cohesiveness. To ensure social sustainability, people and organizations must examine how to promote healthy social interactions and encourage long-term social systems that promote peace in society (German et al. 2022a; Talan et al. 2020). These domains are crucial because they offer a framework for comprehending how individuals behave concerning sustainability. On the other hand, human aspects include things that improve people's quality of life, such as social justice, education, and health.

Studies focusing on human sustainability examined the best ways for people and institutions to build equitable and environmentally friendly communities (Abusafieh and Razem 2017). Moreover, when it comes to determining productivity, one must look at how effectively and efficiently one can generate things and services. This aspect is important since it may help decrease waste and maximize resource consumption by enhancing production, which can enhance the overall performance of both people and organizations (Abdel-Shafy and Mansour 2018). On the other hand, environmental aspects pertain to the ecologically responsible and sustainable practices that are assessed, which also protect various aspects of life (Gansser and Reich 2023). Environmental sustainability encompasses examining methods by which people and companies can preserve resources, lessen their

Table 1 Summarized key related studies and gaps.

Reference	Objective(s)	Method(s)	Main findings	Gap
Park and Lin (2020)	Explored on purchasing experience and the intention of consumers among upcycled and recycled fashion products.	Descriptive analysis, chi-square, and logistic regression	Recycled and upcycled apparel products exhibited a larger purchase gap, resulting in more than 35% of respondents not purchasing these products despite having favorable and strong purchase intentions.	Antecedent of behavioral factors should be explored more with other techniques. There are also still limited studies on the purchasing intention and overall behavior, which needs further assessment.
Zhang et al. (2021)	Analysis of consumer attitude towards sustainability of fast fashion products in the UK.	Correlation and Cross-Tabulation	EU countries' consumers choose to buy from businesses they believe are contributing to social or environmental welfare. Consumers who buy sustainably produced goods reported feeling more content and indicated they would actively choose brands if they were more transparent about their sustainable credentials in both their advertisements and their packaging.	They have only considered smaller sample size due to the COVID-19 pandemic, only students were evaluated, and suggested to use higher prowess analysis tool since only correlation and cross-tabulation was considered. Despite that, correlation could suffice the findings of the study, but could benefit with more data and more recent analytical tools.
Chi et al. (2021)	Examined female millennial consumers' value perceptions for purchasing recycled polyester-made athleisure apparel.	Qualitative analysis: intensive interview process	US millennials considered sustainability as one of the most crucial considerations to take into account when purchasing clothing, wherein the studies showed that women constituted the predominant segment of athleisure consumers and played a leading role in advocating for a sustainable athleisure lifestyle.	Only considered students and only used qualitative exploratory methodology. The study suggested to consider qualitative assessment in the future for related studies to holistically assess sustainable apparel purchase.
Jaheer Mukthar et al. (2024)	Aimed to investigate the impact of product-related factors, personal factors, promotional factors, social factors, and environmental and economic factors on recycled clothing buying behavior.	Structural Equation Modeling	Promotional factors prompted as the most significant variable, followed by personal, social, and environmental factors, while the product-related factors did not have any significance.	Larger sample size (>400 respondents) and alternative methods were suggested to be extended to understand more the recycling buying behavior of consumers. Exploration of more factors was also suggested to be assessed.
Jin et al. (2024)	Combined the customer perceived value and knowledge-attitude-behavior framework to assess both consumer-specific and product-specific factors on recycled products.	Structural Equation Modeling	Four dimensions of perceived value and environmental knowledge as significant factors that positively influenced customers' attitudes toward purchasing recycled apparel. Notably, it was found that perceived quality value had a more pronounced effect on purchase intention than other factors. Interestingly, product knowledge did not demonstrate a significant impact on attitudes.	Expand and implicate more factors in the framework, continuous study due to the progression of the apparel industry, focused only on Chinese consumers, and assessment of broader sustainability factors with other tools.
Pranta et al. (2024)	Examined how consumer behavior and purchasing pattern are affected by recycled cotton-polyester blended clothing, with a particular emphasis on sustainable apparel merchandising.	Mixed method of group interview as qualitative assessment and cross-tabular assessment for the quantitative aspect.	The study was able to assess how recycled fashion products could be consumed in the future. The results show that although consumers aged 25–45 from higher social strata and with higher educational background are becoming more conscious of and interested in sustainability then children aged below 15 and old individual's aged over 45 years comparatively. There are still several obstacles preventing the mainstream use of recycled cotton-polyester blended clothing.	Investigating other factors to create strategies for retail effectiveness, durability, overall quality, and sustainability in accordance with consumer engagement could be further assessed.

Table 1 (continued)

Reference	Objective(s)	Method(s)	Main findings	Gap
Vishnoi et al. (2025)	Proposed to determine the influence of 'Environmental Knowledge' (EK), 'Altruism' (Atr), 'Consumer Confidence' (CC) and constructs of 'Theory of Planned Behavior' (TPB) like 'Attitude' (Atd), 'Subjective Norm' (Sub) and 'Perceived behavioral control' (Pbhc) on consumers' intention to purchase 'Green Apparel Products' (GAPI)	Structural Equation Modeling	The result of SEM showed that 'Atd', 'Sub', and 'CC' were positively related to GAPI. Also, six hypotheses were formulated testing the moderating role of 'Plnn'. The results established that 'Plnn' moderated the relationship between 'Atd', 'Sub', 'CC', and 'GAPI' significantly. This research provides a novel framework to explore the relationship between the 'EK', 'Atr', and 'CC' and Generation Z consumer's 'GAPI'.	Only Gen Z consumers were assessed, and that different consumers (age, geographic location) may have different results. It was also suggested that additional or other tools may be used to further assess the overall result.
Boyer et al. (2025)	Adopted a social practice framework in which consumer attitudes and beliefs play a relatively small role explaining behavior like purchasing decisions. The study focuses specifically on the practice of shopping for jeans, taking advantage of focus groups with adults in the United Kingdom to elicit detailed explanations of their preferences for new, recycled, or used jeans.	Focus group discussion	Shopping for these different categories of jeans involves distinct social practices composed of variable meanings, materials, and competencies that are accessible to some individuals and inaccessible to others. Consequently, one avenue to increasing the relative frequency of jeans reuse becomes a matter of improving consumers' access to the practice of shopping for used jeans rather than attempting to transform their attitudes or beliefs.	Focused only on UK consumers, and that further assessment is needed since studies correlated attitude and impact on behavior were linear, but their study was not. In addition, broader exploration on the sustainability factors could be considered and assessed, such as perception and preference.
Reviewer Suggested Journals for Review				
Sarkar et al. (2022)	Assessed the effectiveness of a variable production system under emissions control policies was observed.	A dual-channel policy was utilized to find the most profitable managerial decisions. This is a fully emission-controlled variable production system covered by green technology investment.	Every stage of the variable production system was seen to emit less carbon emission, helping to obtain larger subsidies from the government. The study has proven that the proposed system provides 8.76% more profit than the traditional production system. The study showed the analytical results which prove that the total profit of the sustainable supply chain is a convex function of both lead time and variance of the lead time. As different channel selling is crucial to any industry for increasing customer demand, the retailer sells their products by hybrid channel and advertises their products online to make them more famous. Determined the most effective approaches in protecting the multi-biofuel supply chain (SC) network from operational and disruption risk, as well as transportation, and suggested policies for reduce energy consumption and carbon emissions (CBE) cost. Sustainable flexible production model with single-type substitutable product production was considered by imposing a cap-and-tax policy, investing in green technology, and advertising for products.	Multiple parties of the supply chain could be considered in the development of mathematical models, uncertainty issues and variables could be considered as extensions, consider recycling products for assessment. Automation could be used to reduce carbon emission, extension of model was suggested to consider investing continuous or discrete investment, and larger third-party logistics and overall supply chain model.
Sarkar et al. (2023)	The study considered online selling, offline selling, and buy-online-pickup-in-store policies for selling products from retail stores.	Built a characteristic relation between buy-online-pickup-in-store, backorder, and on-hand inventory by giving an exact total profit function instead of the expected profit function through classic optimization techniques.		Managerial uncertainties with broader framework could be assessed in other implications and aspects, correlate output with market demands, and other factors may be assessed.
Sarkar et al. (2024)	Examine the impact of lead time and carbon emissions cost on a supply chain network.	Stochastic mathematical model for total cost optimization		Consider other parameters, optimization techniques, and total supply chain process outcome - developing other extended mathematical models.
Kar et al. (2023)	Carbon emission through cap-and-trade mechanism was determined and optimized.	Cap-and-trade mathematical model development and optimization		

Table 1 (continued)				
Reference	Objective(s)	Method(s)	Main findings	Gap
Mridha and Sarkar (2025)	Carbon emission assessment of sustainable supply chain.	Prospective total cost equations optimization was employed in relation with on-hand inventory and backorders.		Effectivity on real-life scenario, dataset, and actual utilization should be assessed.
This study	Objective This study aimed to assess sustainable behavior determinants for purchasing apparel made from plastic waste extending the established pro-environmental behavior model from the theory of planned behavior—testing the sustainability theory of planned behavior.	Methods Advanced Analytical Tool—Machine Learning Ensemble: Random Forest Classifier and Artificial Neural Network	Created implications for flexible demand and carbon emission among supply chain, presenting how carbon tax is less viable compared to controlled minimal emission. Novelty Holistic assessment for sustainability factors has yet been done. Related studies have only considered a combination of some factors but has no overall holistic assessment. Qualitative and quantitative analysis tools have been considered, but studies have suggested to use more recent analytical assessment. Some have suggested to use machine learning ensemble to help build classification models, have higher accuracy of prediction, and overcome limitations of traditional and multivariate statistical analysis.	Research Gap Sustainability domains such as environmental, social or political, and economic aspects could be posited to affect sustainable behavior among consumers (Lee et al. 2020)—however, this has yet been thoroughly assessed with a benchmark framework and advanced analytical tool.

ecological footprint, and spread environmentally friendly behaviors. Lastly, the economic aspect pertains to financial structures and processes that promote the growth and development of the economy (Waheed et al. 2023). Studies that focus on economic sustainability incorporate understanding ethical financial practices, settling financial obligations, and making socially conscious investments (Sedliačiková et al. 2020).

With the aforementioned notions on the sustainability aspect and TPB, this paper aimed to establish the Sustainability Theory of Planned Behavior (STPB) to assess sustainable behavior determinants for purchasing apparel made from plastic waste extending the model and concept from studies (German et al. 2022a; Abusafieh and Razem 2017; Abdel-Shafy and Mansour 2018; Gansser and Reich 2023; Waheed et al. 2023; Sedliačiková et al. 2020). In addition, the objective of the study was to assess if the developed model could be established in the field of consumer behavior among clothing industries using a machine learning algorithm; similar to the studies of German et al. (2022a) on consumer behavior among transportation intention and Gumasing et al. (2023) on technology intention and adoption. This study aimed to answer several research questions:

1. Can the STPB framework holistically assess sustainable behavior determinants for purchasing apparel made from plastic waste?
2. How can machine learning ensemble be employed to create a classification model for behavioral analysis?
3. How accurate could the model test out the dataset for forecasting and modeling sustainability and behavioral domains?
4. What implications, both theoretical and practical, could be built from the output of the study?
5. How can the study be extended based on the output and limitations?

As a contribution, this study could be beneficial to business industries considering that this can give knowledge on how the customers’ purchase intention can impact sustainability aspects among apparel. The results could provide implications for sustainable practices among apparel industries. This study can also be advantageous for the government as its findings can aid in recognizing customer needs and preferences, as well as encourage sustainable production and consumption to create programs and policies for sustainable apparel that are more effective. The findings can contribute to the community, considering that this can promote environmentally friendly behaviors in order to minimize pollution and preserve natural resources. Customer awareness and knowledge of sustainable fashion could be increased, which might encourage them to buy more sustainable apparel. Lastly, this study can benefit future generations by identifying possible constraints to sustainable fashion using the proposed theory, which could be utilized to inform the development of strategies that will boost consumer demand for and acceptance of sustainable behavior shortly.

Literature review and hypotheses

Research framework and hypothesis build-up. Figure 3 illustrates the established STPB model, the research framework employed in this study, to determine the purchase intentions of Filipino customers while purchasing clothing made of recycled plastic. The STPB framework in this study has been considered as an extension of the pro-environmental planned behavior (PEPB) from a sustainable transportation perspective (German et al. 2022a; Ong et al. 2023). It is an expanded version of the PEPB from TPB (Ajzen 1991), integrating fully all sustainability domains. The model included eight factors: Perceived

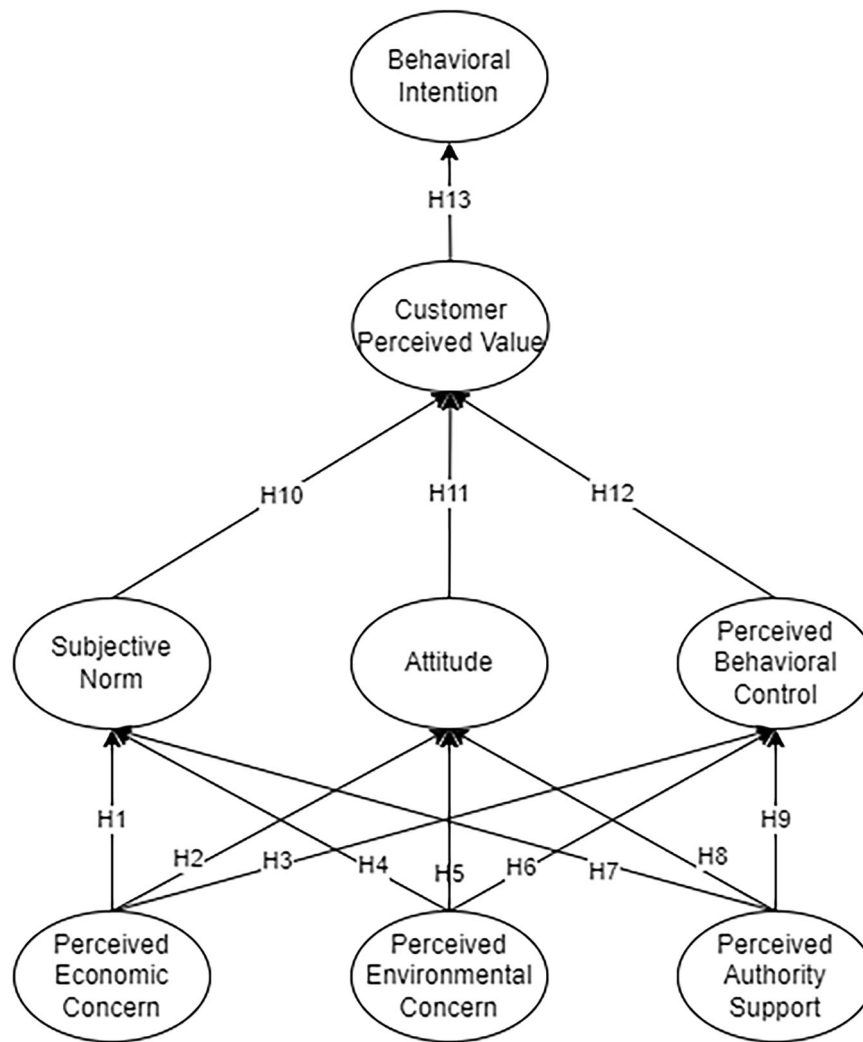


Fig. 3 Theoretical framework.

Environmental Concern (PENC), Perceived Authority Support (PAS), Subjective Norms (SN), Attitude (AT), Perceived Behavioral Control (PBC), Customer Perceived Value (CPV), Behavioral Intention (BI)—PEPB (German et al. 2022a), and the additional Perceived Economic Concern (PECC). As explained in the study of Ong et al. (2023), PECC is an important variable to consider for full sustainability behavioral assessment. The succeeding section provides an outlook of the hypothesis build-up since only PECC is a new addition.

Perceived economic concern (PECC) affecting behavioral domains. PECC can be considered as a gauge of a customer's inclination to allocate additional funds towards sustainable products (Saricam and Okur 2018). Various studies mentioned that customers assert their readiness to buy sustainable clothing, even at a higher price. However, there are uncertainties over whether they do so, making their purchasing behavior contradict their claims (Gomes de Oliveira et al. 2022). Another study also mentioned that consumers who are concerned with environmental issues may not always choose to purchase eco-friendly or sustainable products. Those customers who claim to be concerned with environmental issues might still not engage in pro-environmental behavior upon purchasing products since sustainable products come at a higher cost compared to conventional alternatives (Dangelico et al. 2022), especially evident in the Philippines (Ong et al. 2023). It was mentioned that consumers are frequently willing to invest

more when the real value of a product exceeds their perceived value. However, according to the study of Saricam and Okur (2018), some research efforts aimed to establish the extent to which customers would be willing to pay an extra cost for sustainable fashion products. Moreover, consumers would be ready to spend more on clothing manufactured from organic materials.

The study conducted by Ferioli et al. (2022) and Park and Lin (2020) also indicated that customers exhibit a greater willingness to pay elevated prices for environmentally friendly and sustainable clothing products. To which, a strong positive correlation was found between customers' willingness to spend more (PECC) on sustainable clothing and AT for sustainable apparel (Nam et al. 2017). The study also mentioned that customers with greater environmental concerns tend to show a higher inclination toward purchasing sustainable clothing products. A study conducted by Roh et al. (2022) showed that PECC has a positive value on both SN and PBC, wherein the researchers discovered that the periods when organic products had the most significant development were those in which people were considerably more inclined to change their behavior to promote sustainability. However, Ong et al. (2023) expressed that when too expensive technology is being sold, PECC will not be significant among buying behavior due to price and general economic concerns. It was explained that the more benefit and cost-saving a technology is, the more inclined people will be to purchase. In terms of clothing and apparel, the following were hypothesized:

H1. *Perceived Economic Concern has a significant impact with Subjective Norm.*

H2. *Perceived Economic Concern has a direct significant relationship with Attitude.*

H3. *Perceived Economic Concern has a positive relationship with Perceived Behavioral Control.*

Perceived Environmental Concern (PENC) affecting behavioral domains. PENC can be viewed as a measurement of how each person perceives the effects on the environment (German et al. 2022a; Ong et al. 2023). According to Bickart and Ruth (2012), the degree to which a consumer cares about the environment is a significant personal characteristic because it is linked to their knowledge and motivation regarding environmental matters. Studies have shown that consumers' intentions upon purchasing recycled and upcycled apparel products are positively impacted by environmental concerns (Park and Lin 2020). In a study conducted by Lin et al. (2017), the researchers discovered that PENC has a positive value on both SN and PBC. In contrast, there is little association between PENC and AT in environmental impact assessment (EIA). This correlation shows that customers' PENC was adversely affected when they were required to actively engage in an environmental activity. In the Philippines, it was established that the community is now more inclined to pro-environmental behavior, leading to sustainable options (Ong et al. 2023). Therefore, this study identified the following hypotheses:

H4. *Perceived Environmental Concern has a positive relationship to Subjective Norm.*

H5. *Perceived Environmental Concern has a positive relationship with Attitude.*

H6. *Perceived Environmental Concern has a positive relationship with Perceived Behavioral Control.*

Perceived authority support (PAS) affecting behavioral domains. PAS relates to an individual's comprehension of the resources, laws, regulations, and potentially additional processes provided by a government or authoritative entity to support individuals in adopting a specific behavior (Nadlifatin et al. 2016). According to the study of Lin et al. (2017), PAS positively influences the domains of TPB among citizens' Desire and readiness to engage in an EIA. Considering the viewpoint of AT, these regulations offer regular opportunities for engagement and a variety of communication channels to enhance the positive sentiments of citizens. From the SN perspective, the regulations serve as a means to foster cooperation between the project developer and the broader community. From the PBC perspective, the regulations offer residents the chance to participate in the EIA process under the most convenient conditions (Ong et al. 2023). Therefore, this study identified the following hypotheses:

H7. *Perceived Authority Support has a positive relationship to Subjective Norm.*

H8. *Perceived Authority Support has a positive relationship to Attitude.*

H9. *Perceived Authority Support has a positive relationship with Perceived Behavioral Control.*

TPB domains affecting customer perceived value. AT pertains to the evaluation of an individual concerning the behavior in question, ranging from a positive assessment to a negative one (Soorani and Ahmadvand 2019). On the other hand, SN refers to the perception of societal influence, either encouraging participation in the behavior or discouraging it (Rausch and Kopplin 2021). In other words, SN consists of one's opinions on whether close friends or family members should participate in the behavior. PBC relates to the perception of how easy or challenging it is to perform the activity, encompassing past experiences and

expected obstacles, influenced by one's self-belief and judgment of their capability (Xu et al. 2022). PBC can be utilized to forecast behavior in a straightforward manner as well as indirectly influence behavior through intentions relating to sustainability behaviors (Ong et al. 2023; Soorani and Ahmadvand 2019). In the study of Saricam and Okur (2018), both SN and AT have a significant direct relationship with CPV. Furthermore, the study of Savari and Gharechae (2020), Qi and Ploeger (2019), and Lin et al. (2017) mentioned that TPB domains significantly had a substantial effect on customers' purchasing intentions and played a significant role in influencing them. Behavioral domains in the Philippine context have been established by German et al. (2022a) to affect CPV. Therefore, this study identified the following hypotheses:

H10. *Subjective Norm has a positive relationship to Customer Perceived Value.*

H11. *Attitude has a direct significant relationship to Customer Perceived Value.*

H12. *Perceived Behavioral Control has a positive relationship to Customer Perceived Value.*

Customer perceived value (CPV) affecting behavioral intentions. In this study, CPV illustrates the customer's holistic evaluation of a product's utility, determined by their perception of what has been provided and what they have received (Uzir et al. 2021). It is the outcome of how consumers feel prior to, throughout, and after a purchase has been made (Al-Mashraie et al. 2020; Savari and Gharechae 2020). A study by Dangelico et al. (2022) discovered that CPV is the best indicator of consumers' intentions to purchase sustainable apparel and their willingness to spend higher prices for it, no matter whether eco-material is explicitly utilized. This shows that in the general framework of sustainable clothing, elevated CPV resulting from a product made with a particular eco-friendly material enhances customers' inclination to buy the product, even at a higher price. The study findings of Dangelico et al. (2022) aligned with findings from prior research on sustainable clothing products (Chi et al. 2021). Numerous research investigations have demonstrated that CPV exerts a significant and favorable impact on BI (Jalil et al. 2016; Liu et al. 2021). Therefore, this study hypothesized that:

H13. *Customer Perceived Value has a positive relationship to Behavioral Intention.*

Machine learning algorithm as analytical tool. Recent advancements in artificial intelligence (AI), big data, and machine learning brought newly adapted methodologies for analysis. Ong et al. (2023) explained that the application of machine learning as an analysis tool in behavioral intention among smart transportation provided better output compared to the multivariate analysis counterpart study. This was because several path analyses were present on the large model, creating a total of 18 hypotheses in their study. In accordance, the study of German et al. (2022a) that considered PEPB presented better accuracy for the nonlinear relationship framework established with machine learning technique analyses. From their study, a total of 20 hypotheses were considered in their study. Compared to their other study utilizing higher-order structural equation modeling (SEM) analyses, both generated similar output and could prove how machine learning as another tool could be considered.

When dealing with multivariate analyses, studies such as that of Fan et al. (2016) explained that the larger the framework, the more path is needed to assess the target output. This usually results in lower accuracy of relationship assessment due to errors in the multiple paths needed to be met. Woody (2011) with the same explanation posited that farther variables on the target

output may also result in an insignificant relationship. To which, present studies such as that of Jamshidi et al. (2022) and Al-Mashraie et al. (2020) expressed the advantages of using machine learning techniques in analyzing nonlinear relationship frameworks, which are mostly complex in nature. It was indicated that these hybrid tools could present better output, higher accuracy, and better predictive power. In terms of analyzing behavioral intention among technology adoption, it was presented that machine learning had higher accuracy (Al-Mashraie et al. 2020; Gumasing et al. 2023) and provided a better significance level with classification techniques like neural networks and random forest classifier (RFC; Gumasing et al. 2023).

Methodology

Participants. This study assessed customers' intentions and behaviors regarding sustainable apparel. A total of 500 valid responses were collected through an online survey using a convenience sampling approach. The sampling approach was utilized to disseminate the online survey through various social networking sites and pages in order to procure a wide range of respondents, since convenience sampling is a non-probability sampling method where participants voluntarily choose to participate after being informed about the study by the researcher (Stratton 2021). The first two pages/section of the questionnaire contained a short introduction as well as a reference to the Data Privacy Act and approval of the Ethics Committee (FM-RC-22-01-01, approved on March 20, 2023). Followed by the demographic profiling of the respondents. Lastly, the STPB components were then displayed after the demographic profile was completed. Presented in Table 2 are the participant demographic characteristics, collected alongside the measured items.

According to the collected data, it was observed that the majority of participants were women (69.2%), while males made up the remaining portion (30.8%). Regarding the distribution of age groups, individuals aged 18 to 25 comprised the largest segment, making up (55.4%) of the sample, individuals below 18 comprised (14.4%), and those aged 26 to 35 years old constituted around (13%), while the remaining participants were from older age groups. In relation to marital status, a significant proportion identified as single (75.8%), followed by married individuals at a rate of (20.2%), and both separated and widowed (2%). In terms of residential areas, the majority of respondents, constituting (80.6%), reside in urban areas, while (19.4%) hail from rural regions. Regarding employment status, the majority consists of students (59.2%), followed by employed individuals (31%); unemployed individuals make up a smaller percentage at (5%), while self-employed/business owners account for (4.8%). As for educational level, the highest proportion attended college (40.6%), closely followed by those who attended high school/senior high school (37.4%). A significant portion has completed college or obtained graduate degrees (20.4%); lastly, only a small fraction has attended grade school, with an occurrence rate of just (1.6%). For the household size, the majority of participants have households consisting of 3–4 people (45.6%), followed by 5–6 people (26.4%), more than six people (15%), and finally, those with households of 1–2 people (13%). The majority of participants fall below Php 10,000 (30.8%). Additionally, 23.4% fall within Php 10,001–20,000, and 16.2% fall within Php 20,001–30,000. Those with an income above Php 50,000 account for approximately 13.8%, 10.2% had an income between 30,001 and 40,000, and 5.6% were between Php 30,001–40,000 and between Php 40,001–50,000, respectively. Lastly, for the frequency of purchasing sustainable apparel, (33%) buy sustainable apparel every 1–3 months. Additionally, a significant percentage purchase it every 3–5 months (24.8%), followed by those who buy

it every 6–9 months (23%). Lastly, a small portion of participants make sustainable apparel purchases in 10–12 month intervals (19.2%).

Questionnaire. The questionnaire consisted of two (2) parts: demographic information about potential respondents and determinants of the STPB model, adapted from literature reviews. Supplementary materials present the STPB questionnaire, employing a five-point Likert Scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree) to evaluate the various determinants that influence a user's behavior when it comes to purchasing apparel made from recycled plastic materials. The adapted questionnaire for buildup is presented in the Supplementary File.

Machine learning algorithm (MLA). In this research, a machine learning algorithm ensemble (MLE) was employed, including artificial neural network (ANN) and RFC, which were employed to properly assess the latent variables at once. According to the study by Ong et al. (2022), it was mentioned that using a machine learning ensemble was much more efficient in analyzing the aspects influencing human behavior concerning how people utilize technology compared to traditional and multivariate analysis such as SEM. From their study regarding nuclear power plant reopening adoption among consumers, the SEM analysis proved higher significance on the close variable relationship on the target object. Utilizing MLE, they were able to provide insight into how (1) the basic decision tree showed low accuracy rates compared to RFC, and (2) that farther variables were more significant compared to those close to the target output. They were able to prove that classification techniques such as RFC provided better accuracy output since it generates the most probable model every iteration as compared to the random generation when the basic decision tree is used. Their study also justified the explanation presented by Fan et al. (2016) and Woody (2011)—the farther the variable, the little effect it has on the target output significance level. In addition, SEM is limited to smaller frameworks or smaller path analyses for better predictive power.

Moreover, a comparison of different classification techniques was employed by Ong et al. (2024). They were able to present that RFC and ANN outperformed other classification modeling techniques. For example, there was a significant difference between XGBoost and LightGBM compared to other algorithms. The accuracy rate obtained was lower, with higher mean square errors. This delineates that there needs further improvement on other classification techniques before it could be generalized for use in behavioral studies. It could be deduced that RFC overpowered basic decision trees, XGBoost, and LightGBM, even CATBoost, among others. Studies have also proven that these classification techniques using MLE may also overpower even the advanced multivariate tools like SEM and multiple regression analyses (Öztürk and Başar 2022).

It could be seen among recent studies that little to no studies have used the MLE to evaluate the factors affecting consumers' behavioral intention upon buying sustainable apparel or sustainable behavior in general. Most just focused on technology and its acceptance (Ong et al. 2022), health behavior (Gumasing et al. 2023), transportation (German et al. 2022a; Ong et al. 2023), and adoption (Milani et al. 2020) to name a few. Additionally, studies show that in comparison to SEM (German et al. 2022a; Ong et al. 2022; Ong et al. 2023), machine learning algorithms can produce predictions that are more accurate and models that work more effectively (Bossi et al. 2022). Furthermore, both approaches are capable of handling huge numbers of variables and datasets for

Table 2 Respondents demographic profile.

Characteristics	Category	N	%
Gender	Male	154	30.8%
	Female	346	69.2%
	Total	500	100%
Age	Below 18 years old	72	14.4%
	18–25 years old	277	55.4%
	26–35 years old	65	13%
	36–45 years old	45	9%
	46–55 years old	20	4%
	56–65 years old	14	2.8%
	66 years old and above	7	1.4%
	Total	500	100%
Status	Single	379	75.8%
	Married	101	20.2%
	Separated	10	2%
	Widowed	10	2%
	Total	500	100%
Area of residence	Urban	403	80.6%
	Rural	97	19.4%
	Total	500	100%
Employment	Student	296	59.2%
	Unemployed	25	5%
	Employed	155	31%
	Self-Employed/ Business Owner	24	4.8%
	Total	500	100%
Education level	Finished college or graduate degree	102	20.4%
	Attended college	203	40.6%
	Attended high school/ senior high school	187	37.4%
	Attended grade school	8	1.6%
	Total	500	100%
Household size	1–2 people	65	13%
	3–4 people	228	45.6%
	5–6 people	132	26.4%
	Above 6 people	75	15%
	Total	500	100%
Total Monthly Net Income/ Allowance	Less than 10,000	154	30.8%
	10,001–20,000	177	23.4%
	20,001–30,000	81	16.2%
	30,001–40,000	51	10.2%
	40,001–50,000	28	5.6%
	Above 50,000	69	13.8%
	Total	500	100%
Frequency of buying apparels made from recycled plastic materials	At least every 1–3 months	165	33%
	3–5 months	124	24.8%
	6–9 months	155	23%
	10–12 months	96	19.2%
	Total	500	100%

the development and evaluation of complicated theories (Wendorf 2002).

Various kinds of machine learning algorithms exist, with classification tools being commonly utilized for recognizing patterns. A study conducted by Ong et al. (2022) stated how MLAs like RFC, as well as ANN, have become widely recognized in the field of human factors for evaluating human behavior. However, a study conducted by Jamshidi et al. (2020) emphasized that differences of opinion may arise regarding the utilization of artificial intelligence and machine learning depending on how they are applied as their utilization heavily relies on input from users, individual behavior, and interaction among human roles. To this, they responded by saying that data scientists are in charge of the code and that people who write the algorithm should be in

charge of highlighting its positive features and defining its limits. Despite this, numerous studies have raised questions about the efficacy of machine learning when used independently for measurement and prediction, or when combined with other statistical and multivariate techniques in hybrid approaches.

Random forest classifier (RFC). The RFC is among the many machine learning algorithms commonly employed for classification tasks. It is a classification model which takes into account a straightforward algorithm featuring high prediction accuracy. The study of Chen et al. (2019) mentioned the efficiency of RFC in creating superior classification models compared to a standard or basic decision tree, as RFC consistently generates the most accurate tree every iteration. According to related studies (German et al. 2022a; Gumasing et al. 2023; Ong et al. 2022; Ong et al. 2023), RFC could be employed for the categorization of human factors that affect how well an application is used and adapted or how consumers behave. It was shown that RFC is among the most effective tools for analyzing the factors affecting peoples' decisions.

Adapted from the aforementioned studies, several factors were optimized in order to construct the optimal tree when RFC is employed within the Python Integrated Development Environment, Spyder v5.0. Similarly, the sklearn package was integrated. To which, tree depths between 5 and 7 were taken into consideration, as well as criterion factors such as entropy or Gini criteria, training-testing ratios spanning from 60:40 to 90:10, and splitter choices including random or best. Each combination was one of every parameter between tree depth, criterion, splitter, and ratios. This study has therefore analyzed a total of 4800 iterations upon conducting 100 runs per combination.

Artificial neural network (ANN). ANN contains a more intricate computation and algorithm in contrast with other MLAs. ANN comprises neurons and layers that are connected by arcs, which convert input into output by means of an activation function (Abolghasemi et al. 2020). ANNs can analyze nonlinear models and may give more realistic answers to issues that arise in real life, and it is also capable of generating predictions, which are frequently utilized in prediction areas since they produce more significant outcomes compared to traditional methods (Güven and Şimşir, 2020). According to Jamshidi et al. (2022) and Alam et al. (2021), ANNs can be an effective classification approach for examining factors that have a substantial impact on human behavior. Most research endeavors typically commence with ANNs as a starting point before delving into other forms of neural networks like Deep Learning. It was mentioned that deep learning might be taken into consideration if the accuracy and complexity capacity of ANN provides a low output. Nevertheless, when sufficient predictive power is attained, ANN is adequate. In contrast to complexity, general neural networks are regarded as advanced algorithms already (German et al. 2022a; Jamshidi et al. 2022).

In order to categorize elements influencing human behavior, ANN is currently used in combination or hybrid with SEM (Rehman et al. 2022). Research has shown that the intricate computations within this type of machine learning algorithm can yield more precise results, surpassing the capabilities of SEM, which attempts to simulate the transmission of messages among neurons in the brain (Al-Mashraie et al. 2020). In a study by Alam et al. (2021) a SEM-ANN hybrid was taken into account to ascertain the variables influencing users' perceptions of the usefulness of a mental health application. It has been demonstrated that the outcomes of ANN could accurately anticipate elements influencing human behavior. Furthermore, Kalinić et al. (2021) employed ANN to assess customer happiness. They

showed how this kind of MLA can identify components effectively even in the face of dataset noise and can emphasize crucial variables even when nonlinear connections are present.

To create the best model, the ANN parameters were also optimized by using identical data pre-processing methods as RFC with a parameter setting like training and testing ratios. Similarly, studies by Kalinić et al. (2021), Li et al. (2022), and Jang and Xing (2020) mentioned that they explore various methods for activating the hidden layer (tanh, relu, and softmax), and output layer (softmax, sigmoid, swish). In addition, studies by Yousefzadeh et al. (2021), Jena et al. (2020), and Eckle and Schmidt-Hieber (2019) also mentioned that the optimization process also involved considerations of optimizers such as Adam, RMSProp, and SGD. In this study, the ANN algorithm was implemented using the Spyder v5.0—Python Integrated Development Environment in conjunction with Tensorflow, and the Keras package was employed.

Furthermore, the class was set up to encompass 5 indicators that mirrored the dataset's normal distribution, aligning with the 5-point Likert Scale survey responses. In this study, the parameters were derived from an analysis of various literature sources and their combinations. A total of 27,000 iterations were carried out over 150 epochs, involving 10 runs for every combination of three activation functions for the hidden layer, three activation functions for the output layer, and three different optimizers. This encompassed every conceivable combination, starting from 10 nodes and gradually increasing until reaching 100 nodes within the hidden layer.

Data pre-processing and optimization. The coefficient was given a threshold of 0.20, and a 0.05 p -value was necessary for acceptance. It was seen that any correlation coefficient below the set threshold did not present much significance and would disrupt the analysis (Ong et al. 2023). Therefore, the data cleaning process in this study through the feature selection method considered correlation analysis. To which the items were analyzed on the rescaled target output, behavioral intention. All items were deemed significant underwent data aggregation.

Following related studies, a 60% accuracy threshold was considered for this study to employ significance on the relationship (German et al. 2022a). The higher accuracy rate induced better classification modeling for predicting factors affecting human factors and consumer behavior. The Taylor Diagram was then utilized in this study to compare and assess the acceptability of accuracy rates among MLEs used in the study. Gholami et al. (2020) conducted a study that demonstrated the use of the Taylor Diagram in evaluating model performance through its accuracy, standard deviation, and correlation. In the study conducted, it was determined that a Root Mean Square Error (RMSE) value below 20% was within the satisfactory range. The RMSE considered in the Taylor Diagram is the centered RMSE difference between the simulated accuracy output and observed pattern output from the MLE conducted. Additionally, a correlation value exceeding 90% was regarded as being of significant importance.

Marketing 7P's. After the machine learning algorithm has been used and demographic information about the respondents has been collected a marketing strategy based on the Marketing 7Ps was developed. This tool included seven distinct elements, which are Product, Price, Place, Promotion, Process, Physical Evidence, and People. As all aspects of service marketing fall under the umbrella of the 7Ps in the marketing mix, the concept of the 7Ps may be used to reflect the complexity of sustainable clothing for clothing companies (Ho et al. 2022). The Marketing 7Ps would

help companies, government, and even the society. Thus, in order to enhance the competitive edge of marketers, the concept of 7 ps may be more suitable when studying sustainable apparel from a wider perspective.

Results

Random forest classifier (RFC) results. The RFC output was considered with the optimum parameters of gini and best at 90:10 testing and training ratio. With a 92% accuracy rate, this presented a significant difference upon subjecting the results through analysis of variance. The summarized output is presented in Table 3, which is considered depth 6.

Figure 4 represents the optimum tree with RFC. It could be deduced that the parent node identified CPV (X_1) with a value of less than or equal to -0.248 as a determining factor influencing customers' purchase intentions when purchasing apparel made from plastic. Satisfying the parent node would involve considering the Subjective Norm (X_3) with values less than or equal to 2.261. In addition, meeting this requirement would also consider Perceived Economic Concern (X_4), X_3 , and Attitude (X_2), ultimately resulting in an increased likelihood of customers intending to purchase apparel made from plastic. However, if X_1 did not meet the expectations, both PBC (X_0) and X_1 would be taken into account. This would result in significantly increased purchasing intentions among customers buying plastic-made apparel.

Conversely, if the parent node is not met, it will assess X_0 with values less than or equal to 1.081. Meeting this requirement would involve considering X_4 , X_1 , X_2 , and X_4 leading to very high purchasing intentions for apparel made from plastic. If this condition was not met, it would take into account X_1 and X_3 , resulting in significantly elevated purchase intent. If the child node did not meet the criteria, it would consider X_3 and X_4 , resulting in a high level of purchase intent.

Therefore, according to the results, PECC (X_4) is the most important variable that significantly influenced people's AT (X_2), SN (X_3), and PBC (X_0) to have high purchasing intentions. This implies that CPV (X_1) is a highly significant factor affecting purchasing intentions when it comes to purchasing apparel made from plastic. In order to establish a distinct categorization of hidden factors that impact behavioral intentions, the RFC requires supplementary assistance from other machine learning algorithms due to the diverse range of elements that still exist. Chen et al. (2019) adopted various approaches in addition to incorporating outcomes from the RFC in identifying pertinent latent variables.

Artificial neural network (ANN) results. Performing the ANN, two integrated development environments were used to compare the best parameters. Using MATLAB, the ANN output showed a mean square error of 0.75212 for the validation considering the Elu activation function, 1.0533 for Tanh, and 0.9336 for the sigmoid function. On the other hand, the testing mean square error results were 0.26379, 0.20594, and 0.21595, respectively. Considering the parameters on Spyder v5.0, the accuracy rates were 90.20%, 89.60%, and 84.40%. To which, the final ANN considered the Elu function as the best activation function parameter, which was used in both the hidden and output layer run at 150 epochs (Pradhan and Lee 2010). In accordance, the 80% training and 10 validation fold was considered. It was evident that the model (Fig. 5) was deemed acceptable with the R-squared test value being 0.91 at 30 nodes in the hidden layer.

From the results, it could be seen that the input layer was the different variables considered in this study (from STPB). Because of the nonlinear relationship present, several nodes were needed

for the optimum output to be achieved. Upon optimization, 30 nodes were needed to be analyzed in the neural network model using the Elu Activation Function. As presented in Eq. (1) (Nanni et al. 2022), Elu is an advance of the ReLU function where the value lies at [0.1,0.3]. Similar to ReLU, the value of x when positive lies on the negative region, while its y value would be below zero (Nanni et al. 2022).

$$f(x) = x \text{ if } x > 0 \text{ else } \alpha(e^x - 1) \tag{1}$$

The utility of Elu as an activation function has been expressed by related studies to provide a better accuracy rate when dealing with a nonlinear relationship framework. This is because the analysis provides a smooth result, implicating a better accuracy rate when passing through nodes in the hidden layer. The consequence of Elu providing negative values pushes the calculation to batch normalization, thus improving the learning process of the neural network—creating higher accuracy rates and better to be used with multiple paths, and large, nonlinear frameworks (Kim et al. 2020; Xiangyang et al. 2023). However, Elu has its disadvantages; one of which is that it can only be used in the hidden layer and that it is computationally expensive. Compared to the adopted studies, the current study has presented positively high output when Elu was utilized. From other studies, common results were obtained using sigmoid, tanh, and softmax (German et al. 2022a; Gumasing et al. 2023; Ong et al. 2023; Ong et al. 2022). This is because there are a greater number of paths

and a larger nonlinear relationship framework in this study compared to theirs, and Elu could be the best fit for the analysis. To further validate the output, the validation rate was obtained showing over (under)fitting—Fig. 6. For the discussion section to be more coherent, the SHAP package was considered to generate the relative normalized score of importance to rank the significance of each latent variable affecting behavioral intention to consider sustainable apparel. Table 4 displays the importance scores that have been normalized and will be discussed in the following section. From the findings, CPV emerged as the most prominent latent variable, with the TPB domains of PBC, attitude, and subjective norm following in order of significance. Perceived economic concern was also deemed to be significant. However, a low significant level was seen of PENC and PAS (near 60%).

Discussion

Based on the results, CPV stands out as the most crucial factor in determining customers’ purchasing intention for plastic apparel, accounting for 100%. PBC follows closely at 94.7%, while attitude (AT) plays a substantial role with an influence of 87.4%. This demonstrates the significance of CPV in influencing customers’ intentions towards purchasing apparel made of plastic. CPV is crucial in determining how inclined customers are toward purchasing sustainable apparel products. CPV measures how numerous benefits consumers estimate they will obtain from these items concerning the prices involved. Several important aspects are taken into consideration when customers evaluate the perceived value of sustainable apparel products. Brandão and Costa (2021) indicated that a positive perceived value is linked to a favorable attitude, increased social influence, and a sense of empowerment in addressing obstacles responsible for purchasing sustainable fashion. PBC pertains to the idea that customers who perceive the value of sustainable clothing products will likely feel more empowered and confident in their ability to make purchases effectively. The feeling of control arises when individuals believe that their decisions align with what they consider valuable, boosting confidence in their capacity to make choices that promote sustainability. The attitude of customers played a substantial role in influencing the outcome. Customers who perceive value are more inclined towards developing positive attitudes when buying apparel

Table 3 Random forest classifier results (Depth = 6).

Category	60:40	70:30	80:20	90:10
Best				
Entropy	84.41	83.30	84.47	90.94
Standard deviation	1.215	0.894	0.852	1.003
Gini	87.27	82.81	88.47	92.00
Standard deviation	1.021	0.664	0.502	0.000
Random				
Entropy	82.87	80.51	83.78	85.82
Standard deviation	4.003	4.515	4.967	5.590
Gini	82.61	81.03	84.94	85.30
Standard deviation	3.805	3.991	4.886	5.888

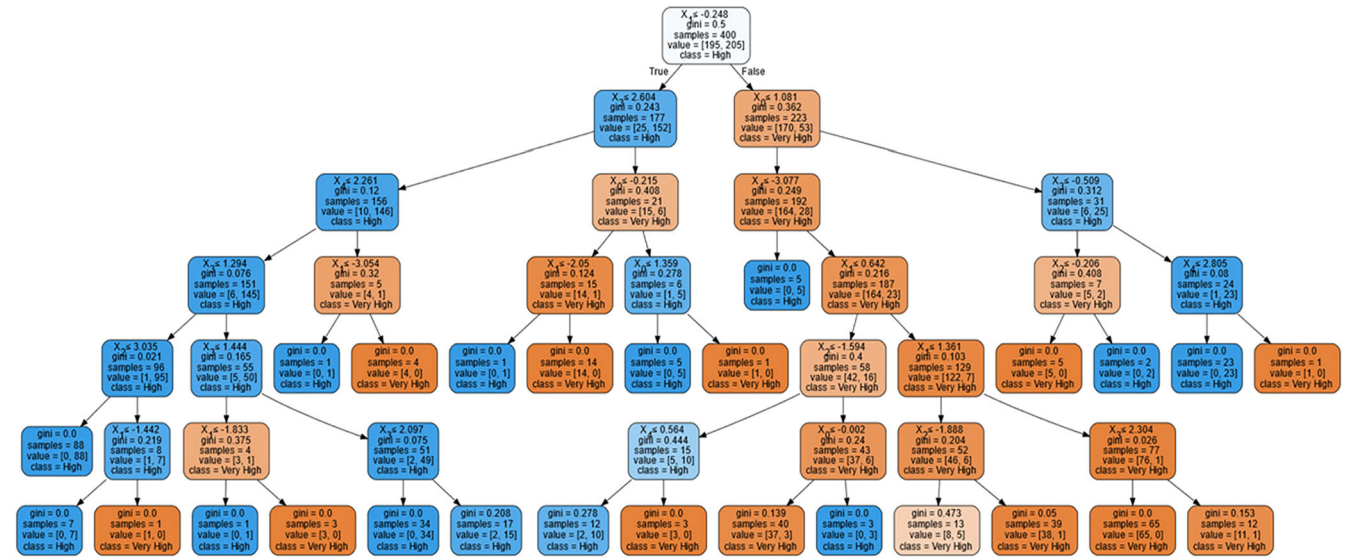


Fig. 4 Optimum classification model with RFC. X_0 - Perceived Behavioral Control; X_1 - Customer Perceived Value; X_2 - Attitude; X_3 - Subjective Norm; X_4 - Perceived Economic Concern.

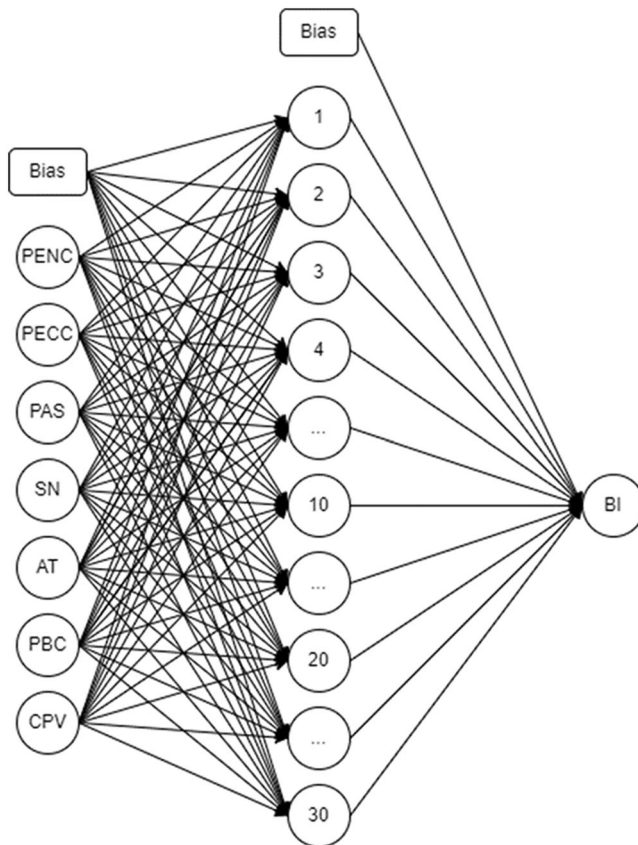


Fig. 5 Optimum ANN classification model.

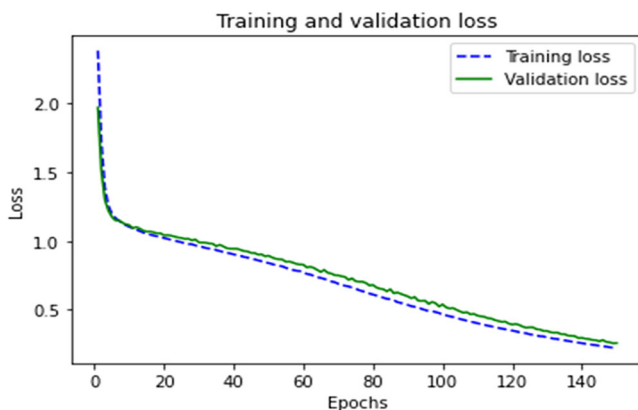


Fig. 6 Validation loss rate.

products made from plastic materials. They perceive these products as environmentally friendly and consider them their preferred choices for which they would willingly spend extra money on sustainable clothing. Brandão and Costa (2021) mentioned that attitude and perceived value are associated with one another. The effects of these actions have an impact on how consumers behave. Therefore, having a strong understanding of the environment may result in a positive attitude (AT) and more significant customer behavioral control (PBC) over the challenges of achieving sustainable apparel consumption.

SN plays a significant role, accounting for 82.6%. SN have a significant influence on shaping consumer behavior, particularly with social and peer norms strongly affecting individuals' purchasing intention towards buying clothes made from plastic

Table 4 Normalized score of importance.

Latent variable	Importance	Normalized score of importance
CPV	0.234	100%
PBC	0.222	94.7%
AT	0.205	87.4%
SN	0.193	82.6%
PECC	0.189	80.6%
PENC	0.183	78.4%
PAS	0.167	71.5%

materials. Businesses and organizations can leverage the power of social approval, peer influence, and the need for acceptance to promote wider adoption of plastic apparel among consumers—making them actively contemplate integrating such products into their wardrobe choices. According to a study conducted by Zhang et al. (2019), subjective social norms show a direct and positive correlation with pro-environmental actions and have a positive and noteworthy influence on the inclination to buy sustainable clothing.

This leads us to the next important factor, the perceived economic concern (PECC), accounting for 80.6% significance. PECC plays a crucial role in influencing customers' intentions regarding purchasing apparel made of plastic. Essentially, it refers to how customers view and contemplate the financial aspects associated with these products. Customers often evaluate whether selecting apparel made from plastic or other alternatives aligns with their budgetary constraints and monetary priorities. They consider various factors like initial purchase price, ongoing expenses, and potential long-term savings. Therefore, if customers perceive that opting for eco-friendly options such as clothing made from plastics is economically viable and can lead to cost reductions or other financial advantages over time, they are more inclined to foster favorable attitudes regarding the utilization of these products. According to a study by Ansu-Mensah (2021), consumers who prioritize environmental concerns are willing to buy sustainable products without reservation, even if it means paying a higher price. Thus, it can be concluded that customers' intentions of purchasing sustainable products are affected by the cost of sustainable products.

The following factor in the hierarchy is known as PENC, accounting for 78.4% of its importance. PENC refers to customers' understanding and sensitivity towards environmental issues, significantly influencing their choices and actions. Customers with a strong sense of PENC are more likely to express strong intentions to purchase clothing items made from plastic materials. Apparel crafted from plastics is often seen as an effective solution for addressing environmental concerns by these environmentally conscious individuals. This group prioritizes preserving nature and considers reduced usage of plastic-based options as a means to lessen their ecological impact or carbon footprint. Such perception boosts their desire even further, leading them to utilize such products to contribute positively towards favorable environmental outcomes. In a study conducted by Zhang et al. (2019), it was mentioned that previous research has typically found a positive link between PENC and the purchase intention to buy sustainable products. Additionally, the study demonstrated a positive association between PENC and both the inclination to buy sustainable clothing products and the attitude toward purchasing them, demonstrating how concerned individuals are with environmental issues and how eager they are to support attempts to address them.

At 71.5%, the PAS ranks lowest among contributing factors on the list. This indicates that customers show importance in receiving support, recommendations, or endorsements from

authoritative figures or institutions regarding their attitudes and beliefs about purchasing sustainable apparel products. It suggests that they value and trust advice or guidance from sources they perceive as credible and having authority. Governments and environmental organizations, according to the study by Lin and Huang (2012), offer subsidies or promotions to encourage individuals to live sustainably. Additionally, it was suggested that for green consumption to become the norm, both the government and environmental organizations must actively promote it. Furthermore, the favorable connection between the educational attainment of customers and their inclination to buy sustainable clothing shows that governments should work towards creating communities that are not just more educated but also more mindful of the environment by investing in education (Dangelico et al. 2022).

The significance of every factor discussed in this paper lies in their ability to exceed the 60% significant level. The ranking was determined solely based on the results derived from various methods employed during data analysis of a survey completed by participants. Overall, it was concluded that the three most important elements that needed to be emphasized in order to encourage individuals to purchase sustainable apparel were CPV, PBC, and attitude (AT).

Theoretical implications. The TPB framework may be used for evaluating consumers' behavioral intentions when it comes to making sustainable apparel purchases. Based on several studies, the TPB framework can be an effective tool for understanding the variables that affect a person's intention when purchasing apparel made from plastic. By considering these variables, interventions may be created that encourage sustainable behavior and the reduction of pollution resulting from plastic waste by means of plastic recycling and clothing production. CPV was shown to have the most significant influence on how consumers felt they had control over their behavior and how they perceived purchasing products made of plastic, which has numerous important theoretical implications. To begin with, it emphasizes the significant impact of external factors on shaping purchase intentions and pro-environmental attitudes, especially when such factors are related to understanding the worth of the products. This study emphasizes the importance of external social variables in influencing individual decision-making and is consistent with well-known behavioral concepts like the TPB. CPV has a beneficial impact on consumers' intentions regarding buying and a readiness to spend extra on products that support sustainable fashion. CPV may therefore be used to assess company strategies toward perceived sustainability and forecast consumer attitudes regarding sustainable apparel. Consequently, the proposed STPB framework may evaluate individual behavior comprehensively with regard to sustainability and sustainable behavior more holistically than other extended TPB or PEPB frameworks.

Practical and managerial implications. To reduce plastic pollution, evaluating the factors influencing customers' behavioral intentions toward purchasing sustainable apparel is essential. According to the study's findings, Filipinos are willing to spend extra on products that support sustainable fashion and are open to buying clothes made of plastic. The community should consider practical implications regarding people's behavioral intentions toward purchasing sustainable apparel. These implications include initiatives and strategies the fashion industry may take to promote a change and lean towards environmentally friendly clothing. Companies may consider sustainable value propositions. Businesses can develop this to improve the perception of the sustainability of their business models. These propositions should

be in line with consumer values and preferences. From this study, CPV—along with the functionality, consumer perception, brand sustainability, desired values, and motivation as a strategy for selling apparel, from plastic to apparel. Specifically,

Functionality Consumers' perceptions of the usefulness of sustainable clothing influence their decisions regarding their purchase.

Consumer perception: By comprehending consumer perceptions and how they affect purchasing decisions, businesses may create environmentally friendly apparel lines that satisfy demand.

Brand sustainability: The decisions consumers make for purchasing sustainable clothing goods are favorably connected with the importance of fashion brand sustainability.

Desired values: Recognizing what consumers want from environmentally friendly clothes may assist businesses in creating sustainable clothing lines that satisfy demand.

Motivations for purchasing: Understanding customer motivations for choosing eco-friendly clothes may help companies create marketing plans that resonate with their target demographic.

By generating sustainable value propositions, gaining insight into consumer perceptions and desired values, and implementing marketing tactics that resonate with their target market, businesses may raise the perceived value of sustainable apparel among their customers. Other suggestions are presented.

For apparel industries. Given the substantial environmental impact associated with the apparel industry, it is more crucial than ever to use sustainable procedures. Sustainable fashion is not merely a fashion trend but also an essential requirement for the industry. Companies have to adapt to keep up with this perspective as customers take an active role in reducing fashion's negative environmental impacts. Ethical and sustainable fashion has evolved from a trend to an economic imperative for the apparel industry. The apparel sector could promote sustainable clothing by implementing eco-design, supply chain sustainability, consumer education, and waste management strategies. The fashion industry must transition to sustainable manufacturing to solve some of the social and environmental issues that societies are now facing. Fast fashion must be reviewed considering the environmental and social consequences it imposes on society, and more sustainable business strategies are critically important. Sustainable manufacturing techniques may be used through ethical labor practices, especially using materials such as recycled plastic bottles and promoting recycling and reuse. As more fashion customers take up the cause of sustainability, apparel manufacturers can advance and profit from the opportunity.

For the government. Governments can play a significant role in advancing sustainable clothing by establishing rules, implementing legislation, encouraging international collaboration, and implementing policy interventions. Regulation may encourage ethical, circular, and sustainable fashion, enabling businesses to view a product's worth as slowly lowering and boosting a circular economy. Laws can promote sustainability in the textile and fashion sectors by compelling companies to provide information on their sustainability development and charging fines if they fall short of goals. International collaboration may support initiatives to change the apparel industry to one that is more sustainable. Sustainable fashion consumption can be influenced by policy changes, such as tax incentives for businesses that employ recyclable materials or provide apparel repair services.

The results imply that companies, environmental advocacy organizations, and political leaders should consider utilizing authoritative support to promote sustainable clothing. This may include pursuing partnerships or endorsements from reputable

people or organizations, displaying certificates or stamps of approval, or actively advertising the potential compatibility of their goods with reliable recommendations. By doing this, companies can increase consumers' positive views and confidence regarding purchasing environmentally friendly products, such as apparel made of plastic, which can encourage more people to engage in environmentally friendly behaviors.

Conclusion

This study explored the factors that impact individuals' behavioral intentions towards purchasing apparel made from plastic. The research utilized two machine learning methods, specifically RFC and ANN and were employed to analyze the collected data. The findings from this research indicated that CPV and PBC were the main variables influencing consumers' behavioral intention toward purchasing apparel made of plastic. Customers who perceive value are more likely to establish positive attitudes (AT) while purchasing sustainable apparel products. The feeling of control arises when individuals believe that their decisions align with what they consider valuable, boosting confidence in their capacity to make choices that promote sustainability. The attitude of customers played a substantial role in influencing the outcome. Customers who perceive value are more inclined to develop positive attitudes when buying apparel products made from plastic materials. They perceive these products as environmentally friendly and consider them their preferred choices for which they would willingly spend extra money on sustainable clothing.

Moreover, customers' inclination towards purchasing sustainable clothing is directly and positively influenced by social norms (SN), which also have a positive and substantial effect on other pro-environmental behaviors. Customer intentions were also affected by perceived economic concern (PECC) because monetary factors were taken into consideration when making purchasing decisions towards sustainable products. Customers with high environmental awareness were more inclined towards purchasing sustainable products as a strategy to reduce their environmental impact and promote environmental preservation, which increases the importance of PENC. Apparel crafted from plastics is often seen as an effective solution for addressing environmental concerns by these environmentally conscious individuals. This group prioritizes preserving nature and considers reduced usage of plastic-based options as a means to lessen their ecological impact or carbon footprint. Such perception boosts their desire even further, leading them to utilize such products to contribute positively towards favorable environmental outcomes.

Finally, PAS refers to how important customers find it to have their attitudes and ideas about purchasing sustainable clothing products supported by or endorsed by reputable individuals or institutions. It was suggested that for green consumption to become the norm, both the government and environmental organizations must actively promote it. Furthermore, the favorable connection between the educational attainment of customers and their inclination to buy sustainable clothing shows that governments should work towards creating communities that are not just more educated but also more mindful of the environment by investing in education.

In summary, it was found that every single one of the aforementioned variables was significant and had a weight of more than 60%. The study emphasized the crucial roles played by perceived value, behavioral control, consumer attitude, economic considerations, environmental concerns, and social norms in determining sustainable consumption behavior. Additionally, it provided important details on the complex interactions between these variables that affect customers' behavioral intentions

regarding purchasing sustainable apparel. This study highlights how important it is to take these factors into consideration when establishing marketing strategies and initiatives intended to raise awareness of products made with plastic in order to make a positive contribution towards the advancement of a sustainable environment to reduce plastic pollution. Furthermore, it opens up the opportunity to conduct further investigation and examination of these factors by employing advanced machine-learning methodologies and larger sets of data to enhance our understanding of sustainable consumer behavior. For ANN, either MATLAB or Python codes could be utilized and could still provide similar findings.

Limitations and future research. The study could be used and expanded to evaluate the sustainability of the apparel industry in various nations as well as the customers' behavioral intentions toward purchasing sustainable apparel. It does, however, have certain restrictions. First, in terms of demographics, the largest portion of survey participants (55.4%) were within the age range of 18 to 25. The research made an effort to include people of all ages; however, because social media platforms were used to gather data, the majority of participants were under 30. In addition, since respondents were simply asked to categorize their region of residence as rural or urban, the study was unable to determine the exact places of residence of the respondents. The specific coordinates of the respondents' locations may have affected the relative importance of various elements, which led to a different view of customer purchasing behavior. Future researchers might improve this by utilizing a more diverse sample procedure and ensuring that various demographic characteristics are considered in their studies. In accordance, real-life testing of model and results could be developed when pre- and post-purchase evaluation and data collection is employed. This could create a better predictive model.

Moreover, the study was unable to take into account and distinguish between online and in-person purchases of sustainable apparel. This could enhance the total scope and depth since it would need to explore deeper into the analysis of a person's intentions when purchasing a particular service or product. Second, the study was unable to relate the respondent's demographics to their behavior, which would have provided a more comprehensive overview of their behavioral intentions when it comes to purchasing sustainable apparel. Future researchers are, therefore, encouraged to offer new perspectives on the subject matter. The findings, metrics, and questionnaires may be used by future researchers to conduct additional studies and to apply new methods and ideas in order to come up with new perspectives and comprehension regarding customer behavioral intentions toward purchasing sustainable apparel. Lastly, future research could look into the role played by customer satisfaction in environmentally friendly purchasing behaviors by concentrating on consumers who have first-hand experience with sustainable clothes.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Competing interests

The authors declare no competing interests.

Ethical approval

This study was approved by the Mapua University Research Ethics Committees (FM-RC-22-01-01), following proper relevant guidelines curated by the university and the ethical

standard of the 1964 Helsinki Declaration. The data entails no traceable information among respondents and was kept in a secure database. This was approved by Dr. Josephine D. German (committee member) and Dr. Michael N. Young (committee head) on March 20, 2023.

Informed consent

Informed consent was obtained from all subjects through written form, involved in this study (FM-RC-22-02-01) during the data collection process from March 27, 2023—August 2023. In accordance, a confirmation question of approval among respondents were collected as the first question in the online questionnaire to ensure approval. Participants were assured that traceable information would not be collected, response would be anonymous, and their data privacy will be secured.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1057/s41599-025-05205-z>.

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